A Novel Density Based Clustering Approach for Electricity Theft Detection

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***Abstract*—The extensive integration of advanced metering infrastructure (AMI) has transformed the landscape of electricity consumption monitoring, providing an effective avenue to address electricity theft by scrutinizing data obtained from smart meters. This research utilises the AMI data to detect instances of electricity theft by employing a novel unsupervised data-driven clustering technique called Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN). The advantage of HDBSCAN in identifying points that do not conform to any cluster, commonly termed as noise points, becomes particularly pertinent in the context of theft detection. These noise points represent potential anomalies or irregularities in electricity consumption patterns, which are crucial for identifying and investigating suspicious activities. Leveraging HDBSCAN for outlier detection enhances the effectiveness of theft detection mechanisms by enabling the identification of data points that deviate significantly from the expected consumption patterns, thereby enhancing the security and integrity of electricity distribution networks. The proposed methodology requires consumption data from smart meters located at consumers premises only. The metering data undergo pre-processing, which encompasses tasks such as recovering missing and erroneous values, as well as normalization, ensuring the accuracy and reliability of the analysis. Furthermore, in our work, we have integrated Density Based Clustering Validation (DBCV) technique for the purpose of hyperparameter tuning, ensuring optimal performance and robustness of the proposed approach. The method was rigorously tested on two real-world datasets, and the results demonstrated robust performance across various evaluation metrics.**

***Index Terms*—Clustering, AMI, electricity theft, HDBSCAN, non-technical losses.**

I. Introduction

ELECTRICITY stands as a cornerstone of modern society, playing a vital role in driving economic activities and fostering growth. Its importance is underscored by its enormous influence across various sectors, impacting both household budgets and industrial operations significantly. However, a substantial portion of generated power remains underutilized, with a notable segment lost within the power system. Grid losses, encompassing technical and non-technical losses, represent a critical challenge. Technical losses (TL), inherent to power dissipation in transmission and distribution lines, are unavoidable due to physical constraints. On the other hand, non-technical losses (NTL), including electricity theft, pose a significant threat to grid reliability and financial stability. The economic repercussions of electricity theft are profound, with global revenue losses reaching 89.3 billion dollars in 2014 and escalating to 96 billion dollars by 2017 [1]. In response, electricity supply companies often resort to tariff increases across customer segments to offset these losses. Reports indicate that each customer in the United Kingdom bears an additional cost of £30 due to electricity theft [1]. The Financial Express report published in 2019 highlighted the economic effects of NTL in India. According to the report, regulatory dues in India amounted to Rs 1.4 lakh crore by the end of the financial year 2018 [2]. This substantial figure underscores the significant impact of NTL on the economy and the challenges faced in managing losses within the power sector.

The deployment of AMI and the extensive use of smart meters have significantly enhanced the capabilities for detecting electricity theft through the vast amount of consumption data they generate. This wealth of data makes data mining technologies particularly well-suited for electricity theft detection, enabling utilities to leverage advanced analytical techniques to identify anomalies and patterns indicative of theft or irregular energy consumption.

II. Related Work

Electricity theft detection techniques can be broadly classified into two main categories: data-oriented and network-oriented [3]. The distinction lies in the type of information they leverage. Network-oriented approaches depend on network topology and measurements, whereas data-oriented methods make use of consumer-related data such as consumer type, load profile, or energy consumption. Within data-oriented methods, there is a further breakdown into supervised and unsupervised techniques [3].

Supervised methods rely on labeled data, so application of these methods to detect electricity fraud requires predefined categories for fraudulent and genuine cases. In [4], theft is detected using data from smart meters, with various supervised learning algorithms applied to analyze the data. Notably, XG Boost exhibited superior performance compared to other

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classifiers in this investigation. In [5], a deep neural network designed for detecting fraudulent customers is proposed. The methodology entails a combination of a Long Short-Term Memory (LSTM) network and a Multi-Layer Perceptron (MLP) network. A sophisticated neural network structure is presented in [6] for the purpose of theft detection, merging the capabilities of a wide component with those of a deep component. This hybrid model is designed to boost detection accuracy by harnessing the broad knowledge acquired by the wide component and the deep component's proficiency in capturing the cyclic patterns in electricity consumption. An Optimum-Path Forest (OPF) classifier was introduced to effectively detect electricity fraud within distribution systems in [7]. In [8], an innovative Electricity Theft Detector named ETD-ConvLSTM is unveiled, harnessing the power of Convolutional Long Short-Term Memory (ConvLSTM) neural networks. Furthermore, [9] delineates an electricity theft detection strategy based on autoencoders and regression algorithms. Initially, the emphasis is on pinpointing individuals engaged in electricity theft, succeeded by the prediction of potentially pilfered amount of electricity to optimize economic gains. Moreover, [10] introduces a methodology that integrates Support Vector Machine (SVM), optimization techniques, and voltage sensitivity analysis for theft detection.

Unsupervised methods differ from supervised methods in that they do not necessitate labeled datasets for classification purposes. Correlation analysis between data obtained from both the master meter and meters installed at customers premises, with the primary objective of detecting potential instances of theft is carried out in [11]. In [12], a data-driven approach is employed for theft detection, utilizing the Maximum Information Coefficient (MIC) method in conjunction with a clustering technique based on CFSFDP (Clustering by Fast Search and Find of Density Peaks). In [13], the detection of theft is accomplished by employing an approach that combines the k-means clustering algorithm with the Local Outlier Factor (LOF) method. In [14], a theft detection system is introduced, which relies on the Cumulative Sum and Shewhart control chart methods for monitoring the process mean. The mean of customers energy consumption is monitored using the Exponentially Weighted Moving Average (EWMA) control chart in [15] to detect potential dishonest consumers. The approach described in [16] focuses on detecting electricity theft by using a Shewhart chart of variance to monitor the variability within consumption data.

Network-oriented methods leverage insights from network analysis and the fundamental principles governing power system behaviour, eliminating the need for labeled data. The work in [17] introduces a load flow-based technique that utilizes consumer premises meter data along with other network measurements such as voltage and power to identify illicitly connected loads within the distribution system. In [18], a method is introduced to compute average TL and NTL in distribution grids by deriving an equivalent operational impedance (EOI) from load flow solutions using average transformer loads obtained from customers' electric bills. The energy theft detection approach presented in [19] employs a novel dynamic programming algorithm to minimize the number of Field Remote Terminal Units (FRTUs) while satisfying the detection rate constraint. In [20], a dynamic programming algorithm is introduced, that inserts the fewest possible Data Protection Regions (DPRs) into a data center. The algorithm's key feature is its use of the Minimum Covariance Determinant (MCD) method to calculate a range of anomaly rates.

All the aforementioned methods have their inherent limitations. The problems associated with exisiting methods for theft detection are summarized below:

1) Supervised approaches, as mentioned in [4], [5], [6], [7], [8], [9], [10], face significant challenges due to their reliance on extensive labeled datasets of both malicious and benign users. Such data can be difficult to obtain and may not accurately represent the target population. Class imbalance is also common, with legitimate consumers far outnumbering fraudulent ones. Methodologies based on supervised learning are trained on specific fraud patterns and they often struggle with emerging fraud behaviors not represented in the training data, making them susceptible to zero-day attacks [3].
2) Network-oriented methods [17], [18], [19], [20] rely on reliable, high-resolution data, including network topologies and RTU information. Additionally, their implementation is expensive due to the need for additional observer meters, RTUs, and increased communication demands. These methods also require advanced software, which raises development and operational costs. While they avoid large datasets, they depend on high-quality data from sources like RTUs and observer meters [3].
3) The methodologies presented in [11], [12] are unsupervised but necessitates additional observer meters with smart meters, significantly increasing implementation costs. While [12] uses clustering for theft detection, it lacks hyperparameter tuning details, posing challenges for real-world application. A uniform hyperparameter set may not be effective across different datasets, limiting the methodology's generalizability and practical use.
4) The methodology in [13] integrates the LOF with K-means clustering for theft detection. However, LOF is sensitive to its hyperparameter, the number of neighbors, and the authors provide no guidance on how to select this parameter, making the approach challenging to apply effectively. Additionally, the K-means algorithm often fails to produce meaningful cluster assignments even when provided with correct number of clusters using elbow method. Its performance suffers when clusters deviate from being round, equally sized, equally dense, or when noise is present. K-means algorithm inherently assumes that clusters are spherical, well-centred, and uniform, which limits its applicability in real-world datasets.
5) The research presented in [14], [15] utilizes control charts, which, while not requiring labeled datasets, necessitate a substantial historical dataset for estimating control limits. Such extensive historical data is often unavailable to utilities.
6) The methodology implemented in [14], assumes that the consumption dataset follows a normal distribution. This assumption is not valid for many real-world datasets.

To address these issues, we propose an unsupervised, data-driven methodology called HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) [21], [22] for theft detection. HDBSCAN stands as a clustering algorithm specifically engineered to identify clusters of diverse shapes and densities within datasets. In [16], the present authors proposed a methodology for theft detection that employs a Shewhart chart of variance to monitor fluctuations within consumption data. Although this approach serves as an effective tool for detecting theft, particularly large-scale theft, it does have some limitations. In this current work, we have addressed these shortcomings and extended our earlier work in the following ways:

1) The Shewhart chart excels at detecting large-scale theft but struggles with smaller-scale thefts. The proposed approach addresses this limitation, delivering reliable results across theft of all sizes, making it more robust and versatile.
2) In our previous study, we focused on detecting a single type of power theft using a control chart-based detector. However, perpetrators often employ diverse methods to reduce their actual power consumption. The methodology proposed in this work detects five common modes of power theft, providing a more comprehensive and effective solution compared to our earlier approach.
3) A major limitation of the control chart-based detector is its reliance on extensive, anomaly-free historical data to define control limits, which is often impractical. The proposed methodology addresses this issue by detecting theft effectively without such data, making it more practical for real-world use.

HDBSCAN enhances the conventional density-based clustering method, DBSCAN (Density-Based Spatial Clustering of Applications with Noise), by implementing a hierarchical methodology. HDBSCAN also possesses the capability to distinguish points that do not align with any cluster, commonly termed as "noise" points. This feature proves valuable in tasks like anomaly detection, where identification of data points that deviate substantially from the general dataset holds paramount importance. The main contributions of this work are as follows:

1) To the best of our knowledge, this is the first application of HDBSCAN methodology for electricity theft detection.
2) Unlike previous studies [11], [12], our approach eliminates the need for observer meters in designated areas, substantially reducing implementation costs. Instead, it relies solely on meters installed at customers' premises.
3) HDBSCAN offers the advantage of not requiring users to preset the number of clusters and can effectively identify clusters of diverse densities [22]. This stands in contrast to the approach outlined in [13], where K-means clustering is employed as discussed earlier.
4) Hyperparameter tuning of machine learning algorithms poses challenges. In our study, we employ an appropriate measure, the Density-Based Clustering Validation index (DBCV) [23], to validate density-based clusters. Leveraging DBCV for hyperparameter tuning enhances the accuracy and reliability of clustering results generated by HDBSCAN. This overcomes the problems associated with hyperparameter tuning in [12], [13].
5) In contrast to [14], which assumes a normal distribution, our non-parametric approach avoids any statistical distribution assumptions.
6) Our methodology does not necessitate a large historical dataset for implementation, thereby addressing the challenge of obtaining such extensive historical data for theft detection, as seen in [14], [15], [16].
7) Our methodology doesn't rely on labeled datasets for theft detection, thus avoiding the limitations of supervised techniques mentioned in [4], [5], [6], [7], [8], [9], [10]. Furthermore, it doesn't require knowledge of network topology or the installation of extra sensors or hardware, overcoming the constraints associated with network-oriented approaches discussed in [17], [18], [19], [20].
8) We have conducted thorough numerical experiments exploring various scenarios of electricity theft utilizing two real-world datasets. Our study includes comparisons with state-of-the-art theft detection techniques, employing these datasets for validation purposes.

Although our work is the first to leverage HDBSCAN for electricity theft detection, its application represents only a small component of our overall methodology. A key advantage of our method lies in its minimal reliance on infrastructure. It requires neither high-quality nor large volumes of historical data, and minimal hardware resources. Furthermore, it operates effectively without making strong assumptions about the statistical properties of the dataset. HDBSCAN was selected specifically for its capacity to address the inherent complexities of theft detection. Unlike conventional clustering techniques, HDBSCAN is adept at identifying clusters with varying shapes and densities, making it particularly suited for capturing the heterogeneous behaviors typically associated with theft scenarios. Its robustness to noise and adaptability to dynamic data distributions further enhance its practicality for real-world electricity consumption data. Additionally, our approach incorporates an efficient hyperparameter tuning framework that resolves the shortcomings of traditional methods, which often rely on fixed hyperparameters or simple metrics such as Silhouette score. This adaptability ensures that the methodology performs well across datasets with diverse characteristics. Because of these features, before conducting experiments, it was hypothesized that HDBSCAN's unique ability to model complex data structures, combined with our hyperparameter tuning strategy, would result in a more reliable and precise detection of anomalous patterns linked to theft. The results of our experiments, detailed in the comparative analysis, confirmed these expectations, highlighting the proposed approach as a scalable, practical, and innovative solution for electricity theft detection.

The subsequent sections of this paper are structured as follows: In Section III, we introduce the HDBSCAN algorithm for clustering and outlier detection, providing a comprehensive outline of the steps involved in both processes. Section IV presents the theft detection framework employed in this research, based on HDBSCAN. This framework encompasses data preprocessing, precise hyperparameter tuning, and the generation of ranked list of consumers. In Section V, we elaborate on the results of experiments conducted on two real-world datasets for theft detection, accompanied by a comparative analysis of our proposed method against other state-of-the-art methodologies. Lastly, in Section VI, we present the conclusions derived from our proposed work.

III. Methodology

HDBSCAN (Hierarchical DBSCAN) builds upon the DBSCAN algorithm introduced in [21]. In their follow-up work [22], the authors presented a comprehensive framework for cluster analysis and outlier detection, addressing DBSCAN's limitations by introducing a hierarchical structure for clusters. This enhancement allows HDBSCAN to identify clusters and outliers across varying densities, making it a versatile tool for such tasks [24]. Additionally, HDBSCAN can be viewed as a robust single-linkage clustering method with flat cluster extraction capabilities [25]. The different components of the HDBSCAN algorithm are as follows:

1) Transforming the space according to density [24] -[25]: The main objective of this step is to ensure the robustness of the clustering algorithm against noise. The core distance, denoted as $core_k\left(a\right)$ refers to the distance from a given point a to its k^{th} nearest neighbour. To address points with low density (high core distance), a novel distance measure called the mutual reachability distance was introduced. Formally, the mutual reachability distance between any two points a and b $(d_m\left(a,b\right))$ is defined as [24]:

$$d_m\left(a,b\right) = \max\ \left(core_k\left(a\right), core_k\left(b\right), d\left(a,b\right)\right) \quad (1)$$

Where the distance $d\left(a,b\right)$ represents the original metric distance between points a and b. min_*samples* (denoted as "k" for brevity in subsequent discussion) determines the minimum number of points required in a neighbourhood for a point to be classified as a core point [24].

2) Building a Minimum Spanning Tree (MST) [24]: This step involves construction of MST utilizing the mutual reachability distance calculated in the first step.

3) Build the cluster hierarchy and compute cluster list [24]: In this step, we construct the cluster hierarchy and compute cluster list. The cluster hierarchy in HDBSCAN represents clusters formed during the clustering process and their relationships at varying levels of granularity. It is created by iteratively removing edges from the extended MST based on weights, leading to clusters and sub-clusters. A key hyperparameter, *min_cluster_size* (denoted as *m*), defines the minimum size for a cluster. Any cluster having a size smaller than the minimum size is labelled as noise. The cluster list records all identified clusters, including their parent-child relationships, and tracks data point memberships at different hierarchy stages for easy reference.

4) Extract the clusters [24], [25]: The next step involves identifying significant clusters from the HDBSCAN hierarchy and the computed cluster list. To do so, a new measure denoted as λ is introduced to assess cluster stability, where $\lambda = \frac{1}{e}$. Here e represents the weight of a particular edge. Additionally, for each cluster, λ_{birth} and λ_{death} are defined as the λ values when the cluster was initially formed from a split and when it was subsequently split, respectively. For every point within a cluster, λ_p represents the λ value at which the point exited the cluster. The stability of a cluster can be computed using these values as [24]:

$$Stability\left(L\right) = \sum_{p\,\in\,\mathrm{L}} \left(\lambda_p - \lambda_{\text{birth}}\right) \quad (2)$$

Here L denotes the cluster for which stability is being calculated. This parameter is utilized for identification of prominent clusters leveraging the HDBSCAN hierarchy. Along with the details of these prominent clusters, the list of cluster assignments can be generated efficiently for each data point.

5) Calculation of outlier scores [24]: Another significant outcome of the HDBSCAN algorithm is the computation of outlier scores for each data point, referred to as GLOSH (Global–Local Outlier Score from Hierarchies) [22]. While computing a data point's GLOSH is not overly complex, it necessitates additional book keeping during the construction of the HDBSCAN hierarchy. During this process, the weight value of the edge being removed is utilized to represent the current hierarchical level. This weight value, along with the last cluster label, must be recorded for every point at the moment it is marked as noise or assigned the noise label. For a given data point a, $\varepsilon\left(a\right)$ indicates the weight value of the point right before it was labeled as noise, and $\varepsilon_{\max}\left(a\right)$ represents the lowest weight at which its last cluster or any of its subclusters still exist. The outlier score for a point a, denoted as $S\left(a\right)$ is given as [24]:

$$S\left(a\right) = 1 - \left(\frac{\varepsilon_{\max}\left(a\right)}{\varepsilon\left(a\right)}\right) \quad (3)$$

Steps 1 - 4 relate to the clustering process of HDBSCAN, whereas step 5 corresponds to the outlier detection phase. The complete process of HDBSCAN algorithm is summarized in Algorithm 1.

IV. Theft Detection Frame Work

Based on the HDBSCAN methodology discussed earlier we present theft detection framework in this section. Let A be the set of consumers in a specific area, where each consumer is represented by elements $\{a_1, a_2, \ldots, a_i, \ldots, a_C\}$. C denotes the total number of consumers, and M represents the total number of days for which power consumption data is available. Set X represents the true daily load profiles of customers, with elements $\{\boldsymbol{x_1}, \boldsymbol{x_2}, \ldots, \boldsymbol{x_i}, \ldots, \boldsymbol{x_c}\}$. $\boldsymbol{x_i}$ denotes the true daily load profile of the i^{th} consumer, structured as $\boldsymbol{x_i} = [x_{i_1}, x_{i_2}, \ldots, x_{i_t}, \ldots, x_{i_n}]$, where x_{i_t} represents the power consumption of the i^{th} consumer at time instant t, and n signifies the total number of readings. Similarly, set $\widetilde{X}$ represents the

Algorithm 1: Main Steps of HDBSCAN Algorithm.

Input G ▷Input Dataset
Output E ▷ Set of Anomaly score for each point.
$E \leftarrow \emptyset$.
for all $(a, b) \in G$ **do**
  Calculate $d_m(a, b)$ using equation (1).
end for
Construct the extended minimum spanning tree from a weighted graph.
Create the HDBSCAN hierarchy based on the extended minimum spanning tree.
Identify the prominent clusters from the hierarchy.
for all $a \in G$ **do**
  Calculate $S\,(a)$ using equation (3).
  Add $S\,(a)$ to E.
end for
Return E.

reported daily load profiles obtained from meters installed at consumers' premises. Element $\widetilde{\boldsymbol{x}}_{\mathbf{i}}$ of $\widetilde{X}$ denotes the reported daily load profile of the i^{th} consumer, formulated as $\widetilde{\boldsymbol{x}}_{\mathbf{i}} = [\widetilde{x}_{i_1}, \widetilde{x}_{i_2}, \ldots, \widetilde{x}_{i_t}, \ldots, \widetilde{x}_{i_n}]$. The set X and $\widetilde{X}$ are available for each of the M days.

It is to be noted that, in the real life scenario, the consumer consumption data are actually available through AMI environment. The AMI environment can also be used for theft detection. AMI environment comprises of a complex network that includes smart meters, communication systems, and data management platforms, all of which facilitate real-time monitoring and management of electricity usage. Smart meters are installed at consumer locations, enabling two-way communication between utility providers and their customers. This arrangement allows for the continuous flow of electricity consumption data, which is crucial for spotting anomalies that may indicate theft. The key components of AMI encompass smart meters that accurately measure usage, secure communication networks that transmit data, and advanced analytics platforms that process and evaluate this information to uncover irregular patterns.

A. Data Preprocessing

Before applying the HDBSCAN algorithm to detect electricity theft, it's crucial to preprocess the daily load profiles of C consumers over M days. The first step of data pre-processing step is recovery of missing and erroneous data. The smart meter readings gathered in AMI often include missing or inaccurate data, attributed to various factors such as malfunctioning smart meters, transmission errors, poor connections, and system maintenance [6]. So, we need a way to tackle missing and erroneous data. The procedure for recovering missing data in the t^{th} element of the reported daily load profile $\widetilde{\boldsymbol{x}}_{\mathbf{i}}$ is defined as follows [6]:

$$\widetilde{x}_{i_t} = mean(\widetilde{\boldsymbol{x}}_{\mathbf{i}}), \qquad if\ \widetilde{x}_{i_t} = NaN \tag{4}$$

Additionally, erroneous data may be present under certain conditions. Hence, the pre-processing step also aims to recover such data using the "three-sigma rule of thumb" with the following equation [6]:

$$\widetilde{x}_{i_t} = \begin{cases} \frac{\widetilde{x}_{i_{t-1}} + \widetilde{x}_{i_{t+1}}}{2} & \text{if } \widetilde{x}_{i_t} > 3\sigma(\widetilde{\boldsymbol{x}}_{\mathbf{i}}) \\ \widetilde{x}_{i_t} \quad otherwise & \end{cases} \tag{5}$$

The function $\sigma(\widetilde{\boldsymbol{x}}_{\mathbf{i}})$ represents the standard deviation of vector $\widetilde{\boldsymbol{x}}_{\mathbf{i}}$. Following this, each load vector is normalized by dividing it by its maximum value.

B. Hyperparameter Tuning

The key hyperparameters for tuning in HDBSCAN are k and m as previously discussed. While the Silhouette Score is a common metric used for hyperparameter optimization in clustering, it assesses cluster cohesion and separation based on distances without considering noise. This limitation makes it less suitable for density-based techniques, which rely on noise and proximity. In contrast, Density-Based Clustering Validation (DBCV) is particularly effective for density-based clustering algorithms because it accounts for noise and captures cluster shapes through densities rather than distances [23]. Let $|D|$ denote the number of clusters obtained from applying the HDBSCAN algorithm, where D represents the set of clusters. DSPC (Density Separation of a Pair of Clusters) denotes the smallest reachability distance found among the internal nodes within the MST belonging to clusters D_i and D_j respectively [23]. On the other hand, DSC (Density Sparseness of a Cluster) is defined as the highest weight among the edges within the MST specific to cluster D_i [23]. Let H represent the total count of data points being evaluated, including any noise. Then, the Density-Based Clustering Validity (DBCV) of the clustering solution generated by HDBSCAN can be expressed as follows [23]:

$$DBCV(D) = \sum_{i=1}^{|D|} \frac{|D_i|}{H} V_D(D_i) \tag{6}$$

where $V_D(D_i)$ represesnts validity of i^{th} cluster and is defined as follows [23]:

$$V_D(D_i) = \frac{\min\limits_{\substack{1 \leq j \leq l \\ j \neq i}} ((\mathrm{DSPC}(D_i, D_j)) - \mathrm{DSC}(D_i)}{\max\left(\min\limits_{\substack{1 \leq j \leq l \\ j \neq i}} (\mathrm{DSPC}(D_i, D_j)), \mathrm{DSC}(D_i) \right)} \tag{7}$$

In the proposed methodology, we employ grid search and the DBCV score for tuning parameters m and k. We conduct the grid search over the range of m and k values from 2 to 20. By evaluating the DBCV score for each combination of m and k, we select the pair that yields the highest DBCV score.

C. Creating Ranked List of Consumers

In HDBSCAN, noise points are identified as outliers (labeled as -1), representing data that do not fit into any cluster and often indicating anomalies or unusual patterns. In our application, these points may highlight consumers engaged in irregular activities, warranting further investigation. Additionally, HDBSCAN

Algorithm 2: Electricity Theft Detection Algortihm Based on HDBSCAN.

Input A, $\widetilde{X}$
Output Z ▷Ranked list of consumers
Create matrix $\boldsymbol{O} = [\widetilde{\boldsymbol{x}}_{\mathbf{1}}, \widetilde{\boldsymbol{x}}_{\mathbf{2}}, \ldots, \widetilde{\boldsymbol{x}}_{\mathbf{i}}, \ldots, \widetilde{\boldsymbol{x}}_{\mathbf{C}}]^T$
Define range of values for m and k. Let these be represented by J and N.
for all $i \in J$ **do**
  for all $l \in N$ **do**
    Perform HDBSCAN as per Algorithm 1 on $\boldsymbol{O}$.
    Calculate DBCV score as described in Section IV-B.
  end for
end for
Select the value of m and k for which DBCV is optimum.
Using the optimal value of k and m apply HDBSCAN as per Algorithm 1 on $\boldsymbol{O}$.
Filter out consumers labelled as noise. Let these consumers be represented by set Z.
Arrange the consumers in Z in descending order with respect to outlier scores.
Return Z

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assigns outlier scores using (3), with higher scores indicating a greater likelihood of fraudulent activity.

In the proposed theft detection work, we consider C consumers, each represented by their reported daily load profile $\widetilde{\mathbf{x}}_{\mathbf{i}}$. Before applying HDBSCAN, these daily load profiles undergo necessary data pre-processing steps. To utilize HDBSCAN effectively, we first determine appropriate k and m values by hyperparameter tuning process as discussed earlier. Subsequently, we isolate data points labeled as -1 or noise, as they are potential indicators of fraudulent behaviour. This initial step serves as a critical component of our theft detection process, aiding in narrowing down the window of inspection. Following the identification of potential fraud consumers, we proceed to rank them in descending order based on their outlier scores. Consequently, we generate a daily ranked list of consumers using the HDBSCAN algorithm, where the cosnumers placed at top of the list have higher chance of being involved in fraudulent activity. The complete process of theft detection methodology based on HDBSCAN is described on Algorithm 2.

V. Experiments and Results

This section presents the results of our study on the effectiveness of the HDBSCAN algorithm, utilizing two real-world datasets. The first dataset contains electricity consumption data from 114 residential units in Western Massachusetts, recorded with a resolution of 1 minute throughout 2016 [26]. The second dataset includes electric power consumption records from the Los Alamos Public Utility Department (LADPU) in New Mexico, collected from smart meters installed in residential homes in North Mesa, Los Alamos [27]. The readings were taken every fifteen minutes.

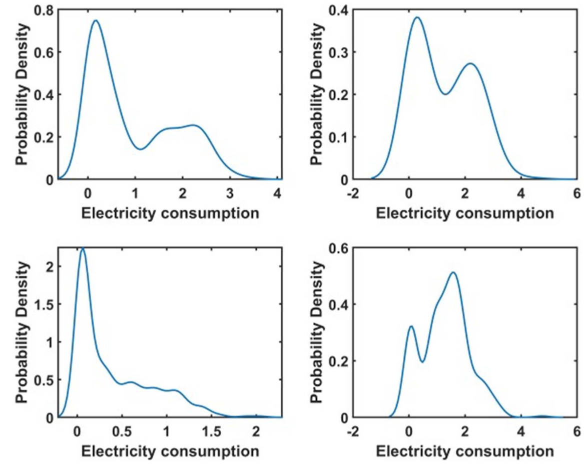

Fig. 1. Statistical distribution of consumer consumption data (Dataset 1).

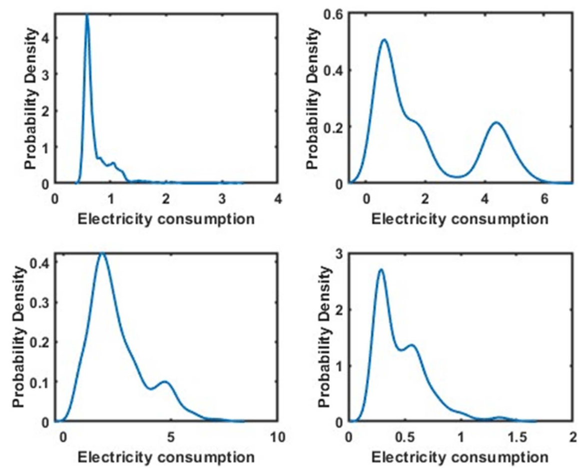

Fig. 2. Statistical distribution of consumer consumption data (Dataset 2).

We selected two real-world datasets to assess the effectiveness of our methodology in real world applications. Figs. 1 and 2 illustrates the statistical distribution of the consumption data for four consumers from the first and second dataset respectively. As shown in these plots, the distribution of the electricity consumption data suggests that the methods relying on normality assumptions are not suitable for real-world scenarios, making these datasets ideal test cases for the non-parametric approach proposed in our work. Furthermore, these datasets have been utilized in previous research to validate theft detection methods, providing a robust foundation for comparing various methodologies against a common benchmark. The rationale for using two datasets is to assess the impact of the unknown inherent arbitrary probability distribution of the consumers dataset on the performance of the proposed method. As would be demonstrated through the results described below, the performance of the proposed method is fairly independent of the inherent probability distribution of the consumer data.

A. Electricity Theft Simulation

In this section, the issue of energy theft is addressed, where individuals tamper with smart meters to lower their electricity bills. It is assumed that the power consumption dataset available to us contain accurate readings. Consequently, we simulated a dataset reflecting false readings for consumers engaged in theft activities. To achieve this, we introduce five distinct theft modes, which will serve as methodologies for converting true consumption values into fraudulent readings. The theft modes (p) are outlined as follows:

1) In the first theft mode ($p = 1$), illegal consumers attempt to decrease their consumption values by a certain percentage. This is achieved by multiplying the actual consumption value at a given time instant by a constant less than 1. This is represented mathematically as [12]:

$$\widetilde{x}_{i_t} = \alpha_t * x_{i_t}, \quad 0.1 < \alpha_t < 0.8 \tag{8}$$

2) In the second theft mode ($p = 2$), consumers engaged in theft activity report scaled-down versions of their daily average readings. This is represented mathematically as [12]:

$$\widetilde{x}_{i_t} = \alpha_t * \bar{x}_i, \quad 0.1 < \alpha_t < 0.8 \tag{9}$$

where $\bar{x}_i$ is the mean value of $\boldsymbol{x_i}$.

3) In Type 3 ($p = 3$) attacks, the smart meter reading is reduced by a randomly selected value which is smaller than the maximum observed consumption value of a particular day. Thus, for $p = 3$ [12]:

$$\widetilde{x}_{i_t} = \max(x_{i_t} - \gamma, 0), \quad \gamma < \max(\boldsymbol{x}_i) \tag{10}$$

4) In a Type 4 ($p = 4$) attack, smart meter readings within a randomly designated time period (t_1, t_2) are replaced with zero values by dishonest consumers. This is represented mathematically as [12]:

$$\widetilde{x}_{i_t} = f(t) \cdot x_{i_t}, \quad f_t = \begin{cases} 0 & \text{if} t_1 < t < t_2 \\ 1 & \text{otherwise} \end{cases} \tag{11}$$

5) This particular theft mode ($p = 5$) is commonly referred to as a mixed mode attack. In this scheme, fraudulent consumers manipulate their load profile by employing a combination of the previous four attack modes [12]. In our study, we illustrate this by sequentially applying theft modes 1 through 4 over multiple days. Such an approach represents the most severe form of attack.

B. Experiments and Results for 1^{st} Dataset

For the first dataset, we analyzed the consumption data from 114 consumers ($C = 114$) for the month of January 2016. This data was originally recorded with a time resolution of one minute, but we converted it to an hourly resolution, resulting in 24 readings per day ($n = 24$). We then divided the 114 consumers into three distinct areas, each containing 38 consumers. In each area, we randomly designated 6 individuals as fraudulent consumers. This newly structured dataset, comprising hourly readings for all 114 consumers, underwent the data pre-processing steps outlined in previous sections.

Initially, we apply the four theft modes as discussed in Section V-A to transform X into $\widetilde{X}$ for each of the 31 days individually. In our final experiment, we simulate a mixed-mode scenario. This scenario involves transforming the consumption data for each day using a different theft mode sequentially. Specifically, theft mode 1 is applied to the consumption data for the first day, followed by theft mode 2 for the second day, and so forth. Once we reach the fourth day and have utilized the fourth attack mode, we loop back to the first attack mode. Consequently, the mixed-mode attack consists of a repetition of all attack modes throughout the 31-day period.

For each mode of theft, we use Algorithm 2 to generate a ranked list of consumers, where those at the top are more likely to be involved in theft. To evaluate the effectiveness of our proposed methodology, we need decision metrics to quantify the quality of the ranked lists our algorithm produces. In our work, we use two well-known decision metrics: Area Under the Curve (AUC) and Mean Average Precision (MAP), to assess the effectiveness of our approach.

The AUC is a frequently used metric for evaluating classification or ranking accuracy [12]. In the context of assessing the ranking performance of an unsupervised method for identifying electricity theft, AUC quantifies the probability that a randomly chosen dishonest customer will hold a higher rank than a randomly chosen normal user. While calculating AUC the consumers are ordered in ascending order of their anomaly scores in the list. The mathematical expression for calculating AUC for the ranked list of consumers is as follows [12]:

$$AUC = \frac{\sum\limits_{i \in F} q_i - \frac{1}{2} * |F| * (|F| + 1)}{|F| * U} \tag{12}$$

In the equation above, q_i represents the rank of the i^{th} user in the list. F denotes the set of actual dishonest consumers present in the list. $|F|$ represents the count of actual dishonest customers, while U denotes the count of normal users in the list. MAP stands for "Mean Average Precision". It is a metric used to evaluate the performance of information retrieval systems. MAP is computed as the average of the precision values for electricity theft detection obtained for each day. Consider V_z to denote the number of fraudulent customers appearing within the top z ranks. Precision at z ($P\,@\,z$) is then defined as [12]:

$$P\,@\,z = \frac{V_z}{z} \tag{13}$$

Then for a given K we can define $AverageP@K$ (denoted as $Avg\,@\,K$) as [12]:

$$Avg\,@\,K = \frac{\sum\limits_{i=1}^{v} P\,@\,z_i}{v} \tag{14}$$

where v is the number of abnormal consumers in the top K positions of the generated list and z_i denote the rank of the i^{th} dishonest consumer within the top K positions of the list. At last we can define $MAP\,@\,K$, which is the average of the daily electricity theft detection precision values, as follows:

$$MAP\,@\,K = \frac{\sum\limits_{d=1}^{M} Avg\,@K\,(d)}{M} \tag{15}$$

TABLE I
RESULTS OF EXPERIMENTS ON 1^{st} DATASET

Theft Mode (p) \ Metrics	Mean AUC	Mean MAP@10
p = 1	0.788 (0.013)	0.680 (0.019)
p = 2	0.847 (0.011)	0.720 (0.022)
p = 3	0.813 (0.018)	0.915 (0.015)
p = 4	0.810 (0.025)	0.890 (0.023)
p = 5	0.820 (0.016)	0.780 (0.018)

where M is the total number of days. We carry out entire set of experiments 30 times for a period of 31 days for each theft mode and then we report the mean value of decision metrics. The results of the experiment conducted on the first dataset, measuring the AUC and MAP@10, are summarized in Table I.Each cell in the table contains the average metric value, followed by its standard deviation in parentheses. These results in Table I reveal the robustness and consistency of the proposed methodology. The findings of the experiments conducted on the first dataset can be summarized as follows:

1) The proposed algorithm achieves high MAP values when the consumers are ranked based on their likelihood of engaging in theft across various theft modes. These high MAP scores across different modes demonstrate the algorithm's capability to consistently rank consumers at the top who have a higher probability of being involved in theft activities. This effectively supports the utility's objective of minimizing the false positive rate.
2) High AUC values across various theft modes demonstrate the algorithm's ability to effectively distinguish between dishonest consumers and legitimate users. The AUC indicates the probability that a dishonest consumer is ranked higher than a legitimate one, demonstrating the model's strong ability to separate classes and identify dishonest consumers. This discriminative ability is key for targeted theft prevention strategies, underscoring the algorithm's reliability in real-world applications.
3) The low standard deviation values for both metrics across theft modes indicate that the methodology's results are quite stable. This stability, combined with consistent performance across all theft modes in terms of both decision metrics, demonstrates that the proposed method is both reliable and efficient.

In our study, it was found that the performance of HDBSCAN in theft mode 1 and 2 is consistent across a broad range of values for the parameter α, specifically between 0.1 and 0.8. This is significant because it indicates that our proposed methodology remains effective across different levels of theft. In contrast, our previous work based on the Shewhart chart of variance, as detailed in [16], only produced desirable results at high levels of theft. HDBSCAN's ability to detect clusters of varying shapes and sizes by adapting to different densities and using a hierarchical approach allows the proposed methodology to effective for different theft modes. This versatility addresses the limitations of previous work in [16], where the performance was only acceptable for a specific type of theft. The results from [16] are not shown here due to space constraints.

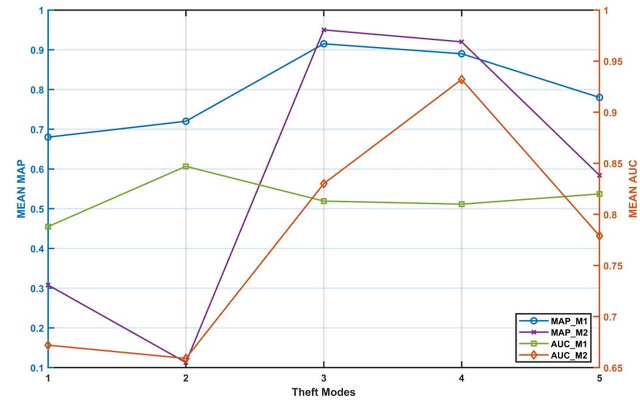

Fig. 3. Variation of decision metrics across all theft modes with and without DBCV tuning.

C. Comparative Analysis of HDBSCAN Performance: With Vs. Without DBCV Tuning

To assess the impact of the DBCV metric for hyperparameter tuning, we conducted two sets of experiments on the first dataset. The first experiment adhered to the procedure outlined in Algorithm 2, applying proper hyperparameter tuning with DBCV as the metric across all theft modes. The resulting metric values from this experiment are denoted as MAP_M1 and AUC_M1. In the second set of experiments, we utilized fixed hyperparameter values (m and $k = 25$) for all theft modes, resulting in metric values denoted as MAP_M2 and AUC_M2. The results of these experiments are presented in Fig. 3.

The plot demonstrates that the HDBSCAN algorithm's performance shows considerable inconsistency across various theft modes when hyperparameter tuning process is not applied. This outcome aligns with expectations, as a particular set of hyperparameters may be optimal for one theft mode but be sub optimal for another. Each theft mode has distinct features and patterns that can significantly impact the algorithm's effectiveness. Therefore, a hyperparameter setting that improves detection in one context might not yield the same results in a different scenario. This highlights the necessity of conducting proper hyperparameter tuning to optimize the algorithm for the specific needs of each theft mode.

D. Validation of the Generalization Ability of Methodology

A key strength of any algorithm is its capacity to generalize effectively. This implies that the method is not limited to specific datasets and can perform well even on data it has not encountered during training. To tackle this concern, we conducted a new series of experiments. The first dataset was split into two segments: 60% for training and hyperparameter selection, and the remaining 40% for testing. Following the selection of hyperparameters based on the 60% training data, we applied the HDBSCAN algorithm to the remaining 40% data using the hyperparameters identified during the training phase. The results of this experiment is presented in Table II.High values of both decision metrics across all theft modes demonstrate that our methodology effectively generalizes to unseen data, confirming that it is not specific to any particular dataset.

TABLE II
GENERALIZATION ASSESSMENT RESULTS FOR THE FIRST DATASET

Theft Mode (p) \ Metrics	Mean AUC	Mean MAP@6
p = 1	0.889 (0.016)	0.910 (0.018)
p = 2	0.920 (0.019)	0.917 (0.012)
p = 3	0.917 (0.012)	0.96 (0.014)
p = 4	0.924 (0.019)	0.96 (0.017)
p = 5	0.915 (0.015)	0.94 (0.021)

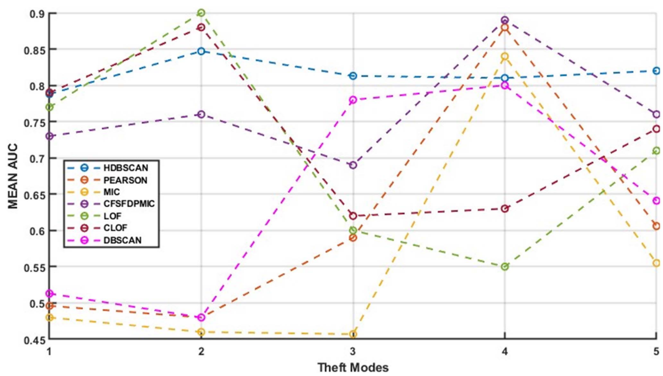

Fig. 4. Mean AUC for different methods across all theft modes.

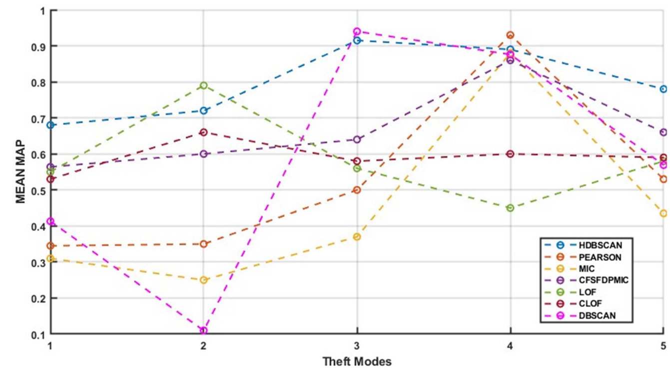

Fig. 5. Mean MAP for different methods across all theft modes.

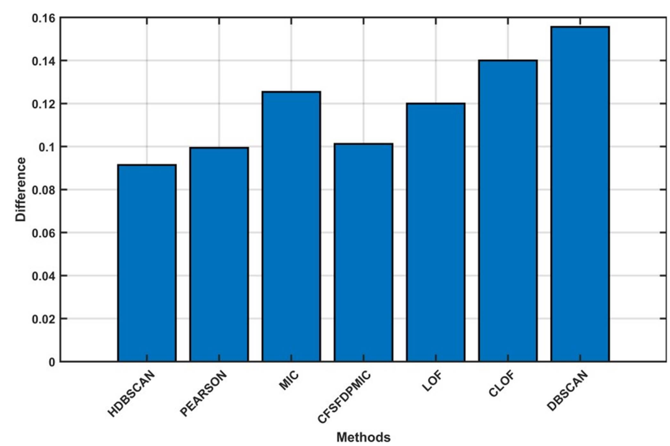

Fig. 6. Difference between both metrics across all theft modes for different methods.

E. Comparison With Other Methods

To showcase the efficacy of the proposed methodology, we have also investigated the performance of additional state of the art correlation analysis techniques and unsupervised outlier detection methods for comparative evaluation. These methods are:

1) Pearson Correlation (PCC): In [11], a methodology based on PCC is introduced for detecting theft.
2) Maximum Information Coefficient [MIC]: This correlation driven methodology has been utilized as a benchmark method in [12] –[13].
3) CFSFDP in combination with MIC (CFSDPMIC): In [12], CFSFDP is utilized alongside MIC for theft detection purposes.
4) LOF: A density-based anomaly detection methodology which is used as a reference method in [11] - [13].
5) Detection based on clustering and LOF (CLOF) : In [13] a methodology which uses K-means algorithm in association with LOF is proposed for theft detection.
6) DBSCAN [28]: A denisty based clustering algorithm applied for theft detction in this work for comparative analysis.

Figs. 4 and 5 show the comparative performance of the proposed methodology against all other methods across all theft modes, as measured by AUC and MAP.

These figures show that the HDBSCAN method consistently outperforms other approaches across all theft modes for both metrics. In contrast, other methods exhibit fluctuating performance, excelling in some modes while underperforming in others. Notably, HDBSCAN's consistent performance is most evident in the mixed mode attack, the most severe form of theft, where it achieves the highest values for both metrics. This success can be attributed to HDBSCAN's ability to adapt to varying densities and detect clusters of different shapes and sizes, making it particularly effective across all theft modes.

Moreover, the difference between metric values across theft modes is notably lower for HDBSCAN compared to all other methods as shown in Fig. 6. It is desirable that a theft detection framework perform well on both metrics. The plot clearly shows that HDBSCAN outperforms the other methodologies in this aspect.

Another notable finding from the comparative analysis is the minimal standard deviation of the decision metric, particularly MAP, achieved by HDBSCAN in comparison to other methods. Fig. 7 illustrates the mean percentage standard deviation of MAP across all theft modes for the evaluated methods. The plot clearly highlights that HDBSCAN consistently provides the most reliable and stable outcomes.

The comprehensive findings underscore that HDBSCAN demonstrates the most consistent, stable, and robust performance across all theft modes, in terms of both decision metrics, when compared to other state-of-the-art theft detection methods. Consequently, HDBSCAN emerges as a very suitable choice for electricity theft detection purposes.

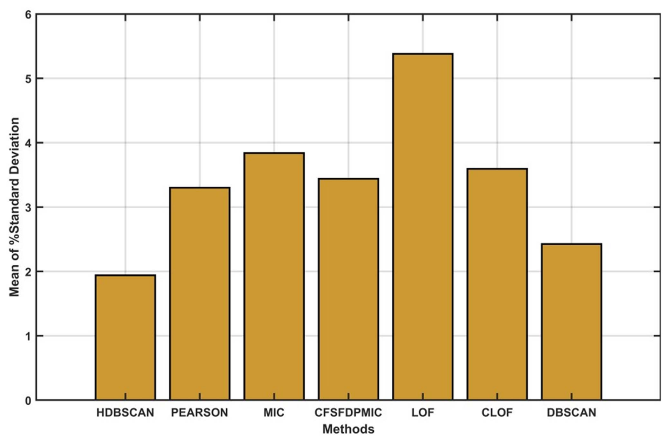

Fig. 7. Mean % standard deviation of MAP across all theft modes for different methods.

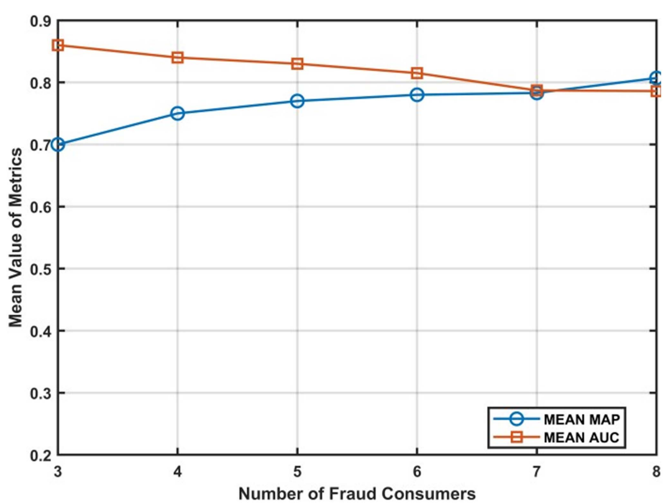

Fig. 8. Variation of decision metrics with change in number of dishonest consumers per area.

F. Sensitivity Analysis

To analyse the sensitivity of the proposed methodology, we conducted two sets of experiments. In the first set, we maintained the number of consumers in each area constant while varying the number of fraudulent users in each area. We adjusted the number of fraudulent consumers from 3 to 8 to observe how the performance of the proposed methodology varied in response to these changes. Fig. 8 displays the results of this experiment. The plot indicates that as the number of dishonest consumers per area increases, there is a slight decrease in the AUC value and a slight increase in the MAP value. However, these changes are not significant or drastic, confirming the stable behaviour of HDBSCAN in response to these variations.

In the second set of experiments, we maintain the number of fraudulent consumers in an area constant and vary the total number of consumers per area. In these experiments, the number of customers per area has been varied from 16 to 57, which consequently alters the total number of areas as well. Fig. 9 illustrates the performance of the HDBSCAN algorithm in terms

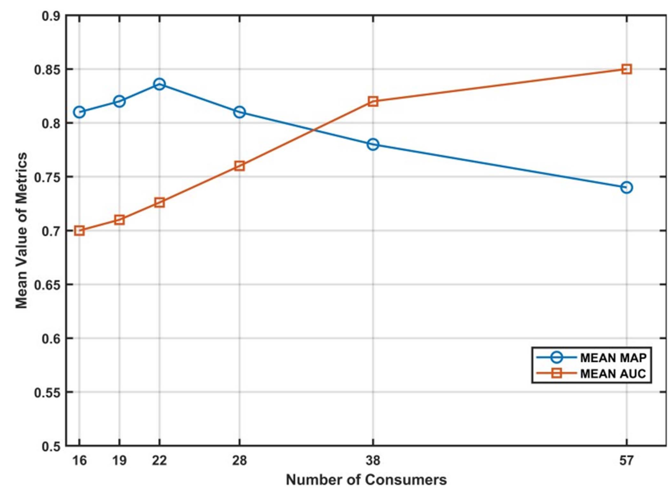

Fig. 9. Variation of decision metrics with change in number of total consumers per area.

of AUC and MAP as the number of consumers per area varies. The plot shows that the AUC value initially increases slowly, followed by a more noticeable rise as the number of consumers grows. However, toward the end, the rate of increase in AUC begins to slow down. In contrast, MAP shows a more subtle trend: it increases slightly at first, then reaches a peak before experiencing a small decline. The changes observed in both AUC and MAP are relatively modest, which underscores the robustness of our approach despite variations in consumer group sizes. The non-linear behavior of both AUC and MAP highlights the dynamics of how the number of consumers influences these performance metrics.

G. Results for 2^{nd} Dataset

In this section, we present the results of experiments conducted on the second dataset to demonstrate the effectiveness of the proposed method across various real-world datasets. As mentioned earlier, the second dataset comprises electric power consumption records sourced from LADPU in New Mexico [27]. Specifically, we utilize the consumption data of 150 households for the month of July 2018. The dataset has a sampling rate of one observation every 15 minutes, which we then convert to a time resolution of 1 h. We divide the total consumers into 3 areas, with each area consisting of 50 consumers. Within each area, we randomly select 9 consumers as fraudulent consumers. Therefore, for each day, we have a set X comprising C consumers. This dataset undergoes data pre-processing steps as discussed in previous sections. Subsequently, we perform the same set of operations as performed on the first dataset to create a ranked list of consumers each day for different theft modes.

The results of the experiment conducted on the second dataset, measuring the AUC and MAP@10, are summarized in Table III. Each cell in the table contains the average metric value, followed by its standard deviation in parentheses. The average values presented in Table III demonstrate that the proposed methodology consistently yields favourable outcomes across all theft modes and decision metrics. The low value of the standard deviations

TABLE III
RESULTS OF EXPERIMENTS ON 2^{nd} DATASET

Theft Mode (p) \ Metrics	Mean AUC	Mean MAP@10
p = 1	0.712 (0.012)	0.64 (0.018)
p = 2	0.825 (0.008)	0.713 (0.027)
p = 3	0.76 (0.013)	0.88 (0.017)
p = 4	0.70(0.019)	0.74 (0.029)
p = 5	0.75 (0.014)	0.74 (0.023)

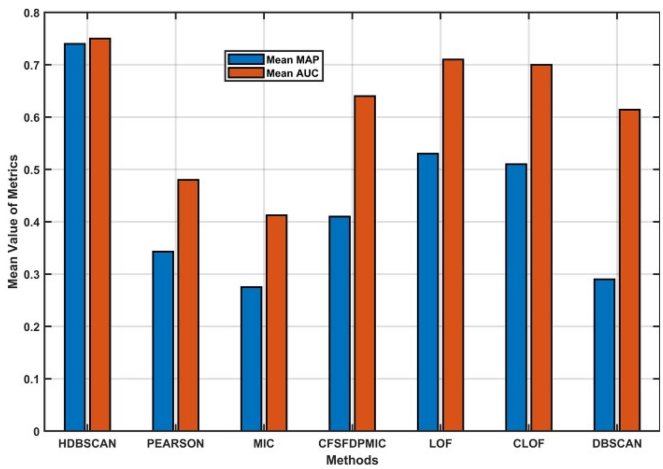

Fig. 10. Mean value of metrics for mixed-mode attack across different methods.

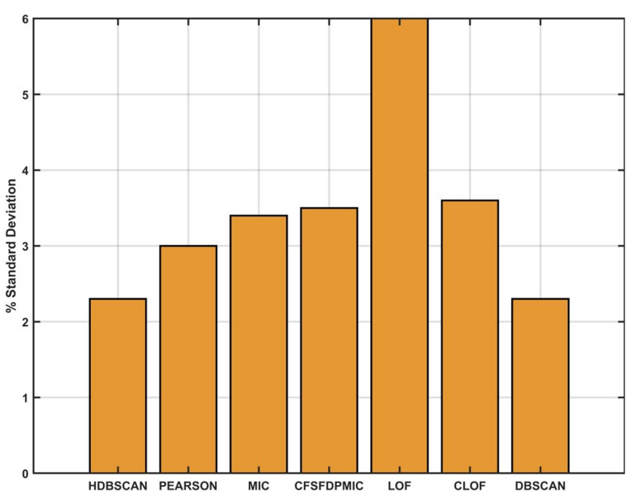

Fig. 11. % standard deviation of MAP for mixed mode attack across different methods.

across all theft modes for both decision metrics further indicate the stability of the results achieved through the application of this methodology. As with the first dataset, we conducted a comparative analysis of the proposed methodology with all other methods on the second dataset. Fig. 10 presents a plot showing the mean values of both decision metrics for mixed mode attack across all methods.

The plot indicates that HDBSCAN outperforms all other methods in terms of both decision metrics on the second dataset as well. Additionally, it's evident that the difference between the metrics for mixed-mode attack is minimum with HDBSCAN as compared to others, which is a desirable indicator of the effectiveness of a theft detection methodology. In Fig. 11, the percentage standard deviation of MAP for mixed mode attack across all methods is depicted. Again, HDBSCAN exhibits the lowest standard deviation compared to other methods. Similar results are obtained for other theft modes as well. This demonstrates HDBSCAN's stability in providing results across all theft modes.

VI. CONCLUSION AND FUTURE SCOPE

A density based clustering methodology, HDBSCAN, is proposed in this work for detecting electricity theft exclusively using smart meter data. The effectiveness of the proposed methodology has been evaluated on real-world datasets using two performance metrics: MAP and AUC. Based on the research carried out in this study, the following conclusions can be drawn:

1) The proposed methodology uses the DBCV method for hyperparameter tuning, effectively overcoming the limitations of traditional metrics such as Silhouette score typically used for this purpose.
2) The results from experiments on real-world datasets show that the methodology consistently performs well across all theft modes with respect to both decision metrics.
3) Compared to existing unsupervised theft detection methods, our approach with HDBSCAN consistently shows superior performance in stability, robustness, and effectiveness across different theft modes and in terms of both decision metrics.

While the HDBSCAN-based approach employed in this work has proven effective in detecting different modes of electricity theft, it presents two key limitations that merit attention. First, the computational complexity of HDBSCAN is significant, posing challenges when applied to large-scale electricity consumption data. With millions of consumer records and extensive time series data, the algorithm's efficiency may diminish, potentially limiting its applicability in very high-volume environments. Second, the methodology does not currently address the growing threat of cyber-attacks on smart meters, which are becoming increasingly vulnerable in the evolving landscape of smart grids. Protecting sensitive customer data is critical as cyber vulnerabilities continue to rise. Future research will focus on exploring less computationally intensive methodologies for theft detection, while also enhancing the algorithm's scalability and incorporating robust cybersecurity measures, such as encryption and anomaly detection systems, to identify both fraudulent activity and potential cyber threats. This will ensure a more comprehensive and resilient solution that addresses both data integrity and security challenges in modern electricity grids.

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